Mathematical and Computational Methods

Exercise Sheet 3: Complex numbers II: exponential form, De Moivre, roots

Tutorial route (▶, in this order, about 80 minutes): 1.3, 2.4, 2.5(a,b), 2.6, 2.8, 2.10(a,b), 2.3. Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 *(⋆ drill — exponential form).* Write each number in exponential form $r\,e^{i\theta}$ with $r > 0$ and $-\pi < \theta \le \pi$:

$$\text{(a) } 1+i, \quad \text{(b) } -3, \quad \text{(c) } 2i, \quad \text{(d) } -1-i, \quad \text{(e) } \sqrt{3}-i, \quad \text{(f) } -2+2\sqrt{3}\,i.$$

Exercise 1.2 *(⋆ drill — back to Cartesian; Euler's identity).* Write each in Cartesian form $a+ib$:

$$\text{(a) } 2\,e^{i\pi/3}, \quad \text{(b) } 3\,e^{i\pi}, \quad \text{(c) } \sqrt{2}\,e^{-i\pi/4}, \quad \text{(d) } 4\,e^{i3\pi/2}, \quad \text{(e) } e^{i5\pi/6}, \quad \text{(f) } e^{i2\pi}.$$

(g) Use (b) to state Euler's identity.

Exercise 1.3 *(▶ ⋆ drill — rotation).* Let $z = 2\,e^{i\pi/6}$. Write down $e^{i\pi/2}z$ in exponential form, and describe geometrically the effect of multiplying a complex number by (i) i, (ii) $e^{i\pi}$, (iii) $e^{-i\pi/2}$.

Exercise 1.4 *(⋆ drill — integer powers).* Compute, using the exponential form:

$$\text{(a) } (1+i)^4, \quad \text{(b) } (\sqrt{3}+i)^3, \quad \text{(c) } (1-i)^6, \quad \text{(d) } \left(2\,e^{i\pi/5}\right)^{10}.$$

(e) Use De Moivre's theorem to simplify $(\cos\theta + i\sin\theta)^{-2}$.

Exercise 1.5 *(⋆ drill — a few roots).* Find all the (a) square roots of i and (b) fourth roots of 1. Plot the roots in (b) on an Argand diagram, mark the unit circle, and name the polygon they form.

Exercise 1.6 *(⋆ drill — the fundamental theorem of algebra).* (a) State the fundamental theorem of algebra, and say what "counted with multiplicity" means. (b) How many roots in $\mathbb{C}$, counted with multiplicity, has (i) a degree-5 polynomial; (ii) $z^7 - 3 = 0$; (iii) $(z-1)^2(z^2+1) = 0$? (c) True or false: every real cubic has at least one real root?

Exercise 1.7 *(⋆ drill — hyperbolic link).* Using $\cos\theta = \tfrac{1}{2}(e^{i\theta} + e^{-i\theta})$, evaluate $\cos(i)$ in terms of e. *(The general identity $\cos(ix) = \cosh x$ is built up in Exercise 2.9.)*

Exercise 1.8 *(⋆ drill — phasors).* (a) Write $5\cos(\omega t + \frac{\pi}{3})$ and $2\sin\omega t$ as phasors $P = A\,e^{i\varphi}$. (b) The phasor $3\,e^{-i\pi/2}$ represents which real signal $A\cos(\omega t + \varphi)$?

2 Tutorial problems

Exercise 2.1 *(⋆⋆ core — arithmetic in exponential form)*. Let $z = \sqrt{3} + \mathrm{i}$ and $w = 1 + \mathrm{i}$. Write z and w in exponential form, then find zw and z/w in *both* exponential and Cartesian form.

Exercise 2.2 *(⋆⋆ core — rotation in action)*. (a) Rotate $z = 3 + 4\mathrm{i}$ anticlockwise by $\frac{\pi}{2}$ about the origin, and then (separately) clockwise by $\frac{\pi}{2}$. (b) Rotate $2 + 2\mathrm{i}$ anticlockwise by $\frac{\pi}{4}$. Give exact Cartesian answers.

Exercise 2.3 *(▶ ⋆⋆ core — prove De Moivre (stretch))*. Prove by induction that $(\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$ for all integers $n \geq 1$. (Pure-maths.)

Exercise 2.4 *(▶ ⋆⋆ core — powers by De Moivre (anchor))*. Compute (a) $(1+\mathrm{i})^8$ and (b) $(\sqrt{3}+\mathrm{i})^6$ using De Moivre's theorem.

Exercise 2.5 *(▶ ⋆⋆ core — roots, two shapes (anchor))*. Find (a) the cube roots of 8 and (b) the fourth roots of -16. In each case give exact Cartesian values and describe the polygon they form on the Argand diagram.

Exercise 2.6 *(▶ ⋆⋆ core — sixth roots of unity (anchor))*. Find all sixth roots of unity in Cartesian form, and verify that they sum to zero. Why does this generalise to all $n \geq 2$?

Exercise 2.7 *(⋆⋆ core — data-science flavour: twiddle factors)*. The fast Fourier transform is built from the "twiddle factors" $W_N^k = \mathrm{e}^{-\mathrm{i}2\pi k/N}$. Take $N = 8$. (a) Write W_8^1 in Cartesian form. (b) Show that $W_8^{k+4} = -W_8^k$ for every integer k (the symmetry that halves the FFT's work). (c) Give the modulus and argument of W_8^3.

Exercise 2.8 *(▶ ⋆⋆ core — $\cos 3\theta$ and $\sin 3\theta$ (anchor))*. Using De Moivre and the binomial theorem, show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ and $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$.

Exercise 2.9 *(⋆⋆ core — trig–hyperbolic link)*. (a) Show $\cos(\mathrm{i}x) = \cosh x$ and $\sin(\mathrm{i}x) = \mathrm{i}\sinh x$. (b) Hence evaluate $\cos(\mathrm{i}\ln 2)$ exactly.

Exercise 2.10 *(▶ ⋆⋆ core — combining oscillations (anchor; physics))*. (a) Combine $3\cos\omega t + 4\sin\omega t$ into a single sinusoid $R\cos(\omega t - \alpha)$. (b) Combine $2\cos\omega t + 2\cos(\omega t + \frac{\pi}{2})$ likewise.

3 Further practice (home)

Roots and the fundamental theorem (Skills 5–6)

Exercise 3.1 *(⋆⋆ core — solving $w^n = z$ with a non-real right-hand side)*. (a) Solve $w^3 = -27\mathrm{i}$, giving the three roots in exponential and Cartesian form. (b) Solve $w^3 = 1 + \mathrm{i}$, leaving the roots in exponential form, and check that each root is the previous one rotated through $\frac{2\pi}{3}$ about the origin.

Exercise 3.2 *(⋆⋆ core — square root the algebraic way)*. The argument of $5 - 12\mathrm{i}$ is not a standard angle, so the exponential method is awkward. Instead find both square roots of $5 - 12\mathrm{i}$ in Cartesian form by equating real and imaginary parts of $(a + \mathrm{i}b)^2$, and verify by squaring.

Exercise 3.3 *(⋆⋆ core — FTA and real factors)*. (a) Verify that $z = 2\mathrm{i}$ is a root of $z^2 + 4$, and write down the other. (b) Factor $z^4 + 4$ into two real quadratics. [Hint: find the four roots first.]

Exercise 3.4 *(⋆ ⋆ ⋆ challenge — a conjugate equation).* Find all $z \in \mathbb{C}$ with $z^2 = \overline{z}$.

Trigonometric identities (Skill 7)

Exercise 3.5 *(⋆⋆ core — power reduction).* Use the inverse Euler formula to write $\sin^4\theta$ as a sum of cosines of multiples of θ.

Exercise 3.6 *(⋆ ⋆ ⋆ challenge — a finite cosine sum).* Using the geometric series for $\sum_{k=0}^{n-1} \mathrm{e}^{\mathrm{i}k\theta}$ (with $\theta \neq 0 \bmod 2\pi$), show that

$$\sum_{k=0}^{n-1} \cos k\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\!\left(\tfrac{(n-1)\theta}{2}\right).$$

Trigonometric–hyperbolic link (Skill 8)

Exercise 3.7 *(⋆⋆ core — the converse pair).* Show that $\cosh(\mathrm{i}x) = \cos x$ and $\sinh(\mathrm{i}x) = \mathrm{i}\sin x$, and evaluate $\sin(2\mathrm{i})$ and $\cos(\pi\mathrm{i})$ in terms of e.

Phasors (Skill 9)

Exercise 3.8 *(⋆⋆ core — physics word problem: two AC sources).* Two alternating voltages of equal amplitude but different phase, $V_1 = 10\cos\omega t$ and $V_2 = 10\cos(\omega t + \frac{2\pi}{3})$ (volts), are added. Find the amplitude and phase of $V_1 + V_2$.